

Start: 2:50 pm

- If research question is clear or reasonable
 - We have a “Gender” column - patients reported → chance that they have used the terms “sex” and “gender” interchangeably so let’s use “sex”
 - Sex is more biologically relevant
 - Overall: good research topic as discussed before - just make sure when you put it in the assignment is well-written
- 1. Does gender modify the relationship between gastric disease histopathological stage and gastric microbiome composition and function?
 - Be clear on what you want to investigate and whether it’s related to the microbiota
 - Sequencing depth range is still within reasonable range (after denoising) (253)
 - 253 bp truncation length is common for 16S rRNA data
 - Choices from curves, including rarefaction, are good choices
 - 3145 rarefaction sampling depth value is good
 - Additional filtering:
 - Up to us if we want to filter out ASVs below 0.5% appearance
 - Should write out why we left them in or took them out
 - Do not filter based on metadata at this step (can do it in R later)
 - Do not need any diversity metrics from QIIME

Next steps:

- Export files into R
- Preliminary analysis (alpha and beta diversity)
- Start thinking about questions in more detail → what will the aims be and what types of statistical tests/metrics we want to use
- PERMANOVA for H.pylori (refer to Meeting 2)
- Look at indicator species for each stage/differential abundance/functional

Aims:

1. Module 13 - alpha/beta diversity
 - a. Comparing the different stages of gastric cancer progress with respect to sex (?)
 - b. Diversity metrics broken down by stages with different between sexes
 - c. Diversity at different stages then stratify it by sex
2. Looking at the taxonomic differences between the stages/groups → indicator species
3. Going into functional analysis (if you see differences, maybe there are:)

- a. Pathways that relates to inflammation
 - b. Sex differences where one sex is doing better
 - c. Metabolism, enzyme, nitty gritty biologic function (kegg terms)
- 4. No random forest